WIN11 OPTIMIZER

PRODUCTION SYSTEM ARCHITECTURE

Document 1 of 3 | Architecture Reference | v1.0

COMPONENT	DESCRIPTION
Scope	Windows 11 | Home / Pro / Enterprise
Layers	Minimal | Moderate | Ultimate | God Mode
Architecture	Layered debloat system — non-destructive first
Goal	Reduce bloat, telemetry, idle resource consumption
Safety Model	Reversible > Irreversible. System integrity preserved.

This document defines the full system architecture of the Windows 11 Optimizer tool — a layered, production-grade debloat system modeled after tools like Chris Titus Tech's WinUtil but engineered with a strict safety-first, reversibility-first design philosophy. Every tweak is categorized, documented, and reversible.

01 | DESIGN PHILOSOPHY

The optimizer is built around one core principle: **never break what you cannot fix.** Unlike random debloat scripts, this system operates in clearly defined layers with a dependency graph, rollback commands, and a state snapshot mechanism.

PRINCIPLE	RULE	RATIONALE
Safety First	No DISM amputations, no core runtime removal	Servicing stack must remain intact
Reversibility	Every tweak has a rollback command	User can undo any layer at any time
Layered Execution	Apply one layer at a time with reboot+observe	Isolates cause of regressions
State Snapshots	Export services/appx/firewall state before any layer	Baseline for diffing and rollback
Transparency	Show exactly what each tweak disables	No blind execution of mystery scripts
Idempotency	Running the same layer twice should be safe	Guards against duplicate operations

02 | SYSTEM LAYER ARCHITECTURE

The system is divided into 4 escalating optimization layers. Each layer includes the previous layer's changes and adds more aggressive modifications. Layers are additive — you cannot apply Ultimate without first applying Minimal and Moderate.

LAYER	CODENAME	RISK	FOCUS	REVERSIBILITY
Layer 1	**MINIMAL**	ZERO	UI/UX noise, basic telemetry, startup	100%
Layer 2	**MODERATE**	LOW	Services, UWP apps, search, SysMain	95%
Layer 3	**ULTIMATE**	MEDIUM	Firewall blocks, scheduled tasks, OEM cleanup	85%
Layer 4	**GOD MODE**	HIGH	Full system hardening, kernel-adjacent services	70%

03 | COMPONENT HIERARCHY MAP

Every optimizable component is mapped to a layer and a category:

CATEGORY	COMPONENT	LAYER	METHOD	RISK
Telemetry	DiagTrack Service	Minimal	sc config	**Zero**
Telemetry	DataCollection Policy	Moderate	Registry/GPO	**Low**
Telemetry	Firewall Endpoint Blocking	Ultimate	WFP Rules	**Medium**
Telemetry	Diagnostic Tasks (CEIP)	God Mode	schtasks	**Low**
Background	UWP Background Access	Minimal	Registry	**Zero**
Background	SysMain (Superfetch)	Moderate	sc config	**Low**
Background	WSearch Indexer	God Mode	sc config	**Medium**

UI Noise	Start Menu Ads	Minimal	Settings	**Zero**
UI Noise	Windows Spotlight	Moderate	Registry	**Zero**
UI Noise	Widgets Infrastructure	God Mode	GPO/Registry	**Low**
UI Noise	Copilot / AI Features	God Mode	Registry	**Low**
Apps (UWP)	Xbox Ecosystem	Moderate	Remove-Appx	**Medium**
Apps (UWP)	Inbox Games / Solitaire	Moderate	Remove-Appx	**Low**
Apps (UWP)	Mixed Reality / 3D Viewer	Moderate	Remove-Appx	**Low**
Services	Xbox Services (4x)	Moderate	Set-Service	**Low**
Services	OneSyncSvc	God Mode	sc config	**Medium**
Services	WMPNetworkSvc	God Mode	sc config	**Low**
Services	Print Spooler	God Mode	sc config	**Low**
Services	MapsBroker	Moderate	sc config	**Zero**
Services	RetailDemo	God Mode	sc config	**Zero**
Performance	Delivery Optimization	Minimal	Settings	**Zero**
Performance	Power Plan Tuning	Minimal	Control Panel	**Zero**
Performance	Pagefile Config	Ultimate	System Props	**Medium**
Performance	Hibernation	Ultimate	powercfg	**Low**
Security	Defender Exclusions	Moderate	Add-MpPref	**Medium**
Security	WER Disable	God Mode	sc config	**Low**
OEM	Vendor App Cleanup	Ultimate	Uninstall-Pkg	**Medium**
OEM	Driver Update Policy	Ultimate	GPO	**Medium**

04 | PROTECTED SYSTEM COMPONENTS

The following components are **NEVER modified** by any layer of this optimizer. These form the immutable core — touching them causes non-serviceable system states.

COMPONENT	BINARY / SERVICE	WHY PROTECTED
NT Kernel	ntoskrnl.exe	Core OS — removing breaks all
Win32 Subsystem	win32k.sys	Graphics / Win32 API layer
Security Authority	lsass.exe	Authentication / Kerberos / SAM
Session Manager	smss.exe	Session init — breaking = no boot
Client/Server RT	csrss.exe	Win32 console subsystem
Logon Process	winlogon.exe	Authentication flow controller
Windows Update	wuauserv / WUService	Servicing stack — patches security
Servicing Stack	TrustedInstaller	Component-based servicing (CBS)
Windows Defender	WinDefend / MsMpSvc	Primary security — never disable
Windows Firewall	mpssvc	Base network security
Store Frameworks	Microsoft.UI.Xaml etc	Runtime for modern apps
RPC Subsystem	RpcSs	IPC backbone — breaking = COM fails
WMI Core	winmgmt	System management interface
Shell Host	ShellExperienceHost	Windows 11 shell rendering

05 | SYSTEM STATE MACHINE

The optimizer operates as a state machine. Each layer transition requires a snapshot, a reboot, an observation window, and a verification step before proceeding to the next layer.

STATE	CONDITION	NEXT ACTION
CLEAN	Stock Windows 11 install	Take snapshot → Apply Minimal
MINIMAL_APPLIED	Layer 1 tweaks active, system stable	Observe 48h → Apply Moderate
MODERATE_APPLIED	Layer 2 tweaks active, system stable	Observe 48h → Apply Ultimate
ULTIMATE_APPLIED	Layer 3 tweaks active, system stable	Observe 72h → Apply God Mode
GODMODE_APPLIED	All layers active, system stable	Run verification suite
REGRESSION	Any layer caused instability	Rollback last layer → re-test
RECOVERY	Rollback failed	Use system restore / disk image

06 | TOOL ARCHITECTURE (TUI Application)

The optimizer is a standalone TUI (Terminal UI) application — not a 'run and pray' PowerShell script. It provides interactive layer selection, live system state feedback, rollback options, and a verification scanner.

MODULE	RESPONSIBILITY	IMPLEMENTATION
Main Menu	Layer selection + system status dashboard	ASCII/TUI with color codes
Snapshot Engine	Export services/appx/tasks/firewall state	PowerShell + CSV/JSON export
Tweak Engine	Execute tweaks per layer with dry-run mode	PowerShell functions
Rollback Engine	Reverse any applied layer or single tweak	Inverse command store
Verification Suite	Confirm system integrity post-tweak	Service + process checker
Logger	Record all changes with timestamps	Append-only log file
Recovery Guide	Step-by-step recovery wizard on error	Interactive TUI panels
Config Store	Persist user preferences per session	JSON config file

07 | PERFORMANCE IMPACT MATRIX

Expected measurable improvements by layer:

METRIC	STOCK	MINIMAL	MODERATE	ULTIMATE	GOD MODE
Idle RAM	~3.2 GB	~2.5 GB	~2.0 GB	~1.8 GB	~1.6 GB
Background Procs	85-100	65-75	50-60	40-50	~35-40
Boot Time	Baseline	-25%	-40%	-55%	-65%
CPU Idle %	5-12%	2-4%	1-3%	0-2%	0-1%
Disk I/O Idle	High	Moderate	Low	Very Low	Minimal
Telemetry Traffic	Full	Reduced	Minimal	Blocked	Near Zero
Net Background	Moderate	Low	Very Low	Minimal	Near Zero

08 | INTER-LAYER DEPENDENCY RULES

Certain tweaks in higher layers depend on components that lower layers may have modified. This dependency map ensures no conflicts occur during layered application.

TWEAK	DEPENDS ON	CONFLICT RISK
Firewall Telemetry Block	WFP service running	Low — WFP is protected
Remove Xbox Appx	Store runtime present	Low — Store not removed
Disable WSearch	Search not excl. already	Zero — additive
Defender Exclusions	Defender running (always)	Zero — Defender protected
Disable WerSvc	EventLog still active	Low — EventLog protected
Pagefile Adjust	RAM > 8GB recommended	Medium — OOM risk on 4GB
Print Spooler off	No active print jobs	Low — manual re-enable easy
Disable OneSyncSvc	OneDrive already removed	Zero — independent
Disable SysMain	SSD confirmed	Low — matters on HDD

09 | REVERSIBILITY CLASSIFICATION

All tweaks are assigned a reversibility class. This determines the recovery procedure when a rollback is needed.

CLASS	DESCRIPTION	RECOVERY METHOD	EXAMPLES
R1 — Full	Single command reversal	Rollback script	sc config, reg delete
R2 — Easy	Settings toggle or script	Settings UI or script	Edge settings, Power plan
R3 — Mod.	Store reinstall or re-register	Microsoft Store / PS	Xbox apps, Sticky Notes
R4 — Hard	OEM reinstall needed	Vendor website download	OEM utilities, drivers
R5 — Image	System image restore only	Clonezilla / Macrium	Corrupted servicing stack

10 | SNAPSHOT COMMAND REFERENCE

Before applying **any layer**, the tool automatically runs these snapshot commands to preserve the system baseline. These files are stored in C:\WinOptimizer\snapshots\

```
Get-Service | Select Name,DisplayName,Status,StartType | Export-Csv
C:\WinOptimizer\snapshots\services_baseline.csv -NoTypeInformation
```

```
Get-AppxProvisionedPackage -Online | Select DisplayName,PackageName | Export-Csv
C:\WinOptimizer\snapshots\appx_baseline.csv -NoTypeInformation
```

```
schtasks /query /fo LIST > C:\WinOptimizer\snapshots\tasks_baseline.txt
```

```
Get-NetFirewallRule | Select Name,DisplayName,Direction,Action,Enabled | Export-Csv
C:\WinOptimizer\snapshots\firewall_baseline.csv -NoTypeInformation
```

```
reg export HKLM\SOFTWARE\Policies C:\WinOptimizer\snapshots\policies_HKLM.reg /y
```

```
reg export HKCU\SOFTWARE\Policies C:\WinOptimizer\snapshots\policies_HKCU.reg /y
```

```
Enable-ComputerRestore -Drive "C:\"; Checkpoint-Computer -Description
"pre-optimizer-$(Get-Date -f yyyyMMdd)" -RestorePointType MODIFY_SETTINGS
```

11 | COMPLETE TWEAK CATALOG (ALL LAYERS)

[LAYER 1 — MINIMAL (Zero Risk)]

#	TWEAK	METHOD	RISK	REV
1	Disable DiagTrack telemetry service	sc config	Zero	R1
2	Disable tailored experiences + activity history	Settings	Zero	R2
3	Disable background apps globally (UWP)	Registry	Zero	R1
4	Disable startup bloat (Teams, Discord, Spotify...)	Task Manager	Zero	R2
5	Remove Start menu ads + suggestions	Settings	Zero	R2
6	Disable lock screen Windows Spotlight	Settings	Zero	R2
7	Disable Edge startup boost + background extensions	Edge Settings	Zero	R2
8	Disable Widgets via Group Policy	gpedit.msc	Zero	R1
9	Disable Xbox Game Bar	Settings/Registry	Zero	R1
10	Disable Cortana startup	Apps Settings	Zero	R2
11	Disable OneDrive startup (optional)	Startup/Uninstall	Zero	R2
12	File Explorer: This PC + disable recent files	Explorer Options	Zero	R2
13	Disable Windows Tips notifications	Settings	Zero	R2
14	Reduce Search Indexing (exclude dev/data folders)	Settings/Registry	Zero	R2
15	Balanced power plan optimization	Control Panel	Zero	R2
16	Disable Delivery Optimization P2P	Settings	Zero	R2

[LAYER 2 — MODERATE (Low Risk)]

#	TWEAK	METHOD	RISK	REV
1	Telemetry policy to Level 0 via registry (AllowTelemetry)	Registry/GPO	Low	R1
2	Disable Advertising ID	Registry	Zero	R1
3	Disable Xbox services: XblAuthManager, XblGameSave, etc.	Set-Service	Low	R1
4	Disable WMPNetworkSvc, RetailDemo, Fax, MapsBroker	Set-Service	Zero	R1
5	Disable dmwappushservice (device mgmt push)	Set-Service	Low	R1
6	Disable telemetry scheduled tasks (CEIP, AppExperience)	schtasks	Low	R1
7	Remove Xbox, 3D Viewer, Solitaire, GetHelp UWP packages	Remove-Appx	Med	R3
8	Remove Edge scheduled background helper tasks	schtasks	Low	R1
9	Uninstall OneDrive (if not needed)	OneDriveSetup	Low	R1
10	Exclude dev/dataset folders from Search Index	Settings	Low	R2
11	Disable SysMain (Superfetch) — SSD systems only	sc config	Low	R1
12	Add Defender exclusions for datasets, VMs, node_modules	Add-MpPreference	Med	R1
13	Disable registry UI noise (tips, suggestions, Spotlight)	Registry	Zero	R1
14	Uninstall OEM bloatware apps (keep drivers)	Apps & Features	Med	R4

[LAYER 3 — ULTIMATE (Medium Risk)]

#	TWEAK	METHOD	RISK	REV

1	Block telemetry endpoints via WFP firewall rules	New-NetFirewallRule	Med	R1
2	Un-provision AppX for ALL users (Xbox, MixedReality, etc.)	Remove-AppxProv.	Med	R3
3	Disable WSearch indexer service entirely	sc config	Med	R1
4	Disable WerSvc (Windows Error Reporting)	sc config	Low	R1
5	Disable hibernation to recover disk space	powercfg -h off	Low	R1
6	Configure pagefile: fixed size or secondary drive	System Properties	Med	R2
7	Disable driver update distribution via Windows Update	GPO	Med	R2
8	Disable additional telemetry scheduled tasks (deep set)	schtasks	Low	R1
9	OEM utilities cleanup — remove all non-driver vendor apps	Uninstall-Package	Med	R4

[LAYER 4 — GOD MODE (High Risk — Full System Hardening)]

#	TWEAK	METHOD	RISK	REV
1	Disable ContentDeliveryManager consumer features	Registry	Low	R1
2	Disable Windows Spotlight system (CloudContent policy)	Registry	Zero	R1
3	Disable Widgets infrastructure (AllowNewsAndInterests = 0)	Registry/GPO	Low	R1
4	Disable Windows Copilot + Recall AI features	Registry	Low	R1
5	Disable DiagTrack + all CEIP scheduled tasks (full set)	sc + schtasks	Low	R1
6	Disable OneSyncSvc (Microsoft account settings sync)	sc config	Med	R1
7	Disable WMPNetworkSvc + MediaSharing	sc config	Low	R1
8	Disable Print Spooler (if printing never needed)	sc config	Low	R1
9	Disable MapsBroker fully	sc config	Zero	R1
10	Disable RetailDemo service	sc config	Zero	R1
11	Disable WSearch completely + replace with Everything Search	sc config	Med	R1
12	Remove full Xbox ecosystem packages + services	Remove-Appx+svc	Med	R3
13	Disable background app infrastructure globally	Registry	Med	R1
14	Disable Delivery Optimization via registry	Registry	Zero	R1
15	Disable SysMain fully	sc config	Low	R1
16	Disable WerSvc crash reporting	sc config	Low	R1
17	Block additional telemetry IPs via firewall	NetFirewallRule	Med	R1

This architecture document is Document 1 of 3. See Document 2 for implementation strategy and Document 3 for testing and verification procedures.